复旦数学资料汇编
规范场

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第一章　规范场

1.1　习题: 规范场

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1.2　习题: 规范场/Example Sheet 1

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范围包括微分流形、李群、李代数、表示.

Question 1.2.1. *Let M and N be smooth manifolds and $\phi : M \to N$ a smooth map.*

- *Suppose X and Y are vector fields on M and N respectively. If Y is ϕ-related to X (i.e. $Y_{\phi(p)} = (D_p\phi)(X_p)$,$\forall p \in M$), prove that*

$$(L_Y f) \circ \phi = L_X(f \circ \phi), \forall f \in C^\infty(N).$$

- *If X' is ϕ-related to Y, prove that $[X', Y']$ is ϕ-related to $[X, Y]$.*

Solution:
If $Y_{\phi(p)} = (D_p\phi)X_p, \forall p \in M$, then

$$(L_Y f) \circ \phi(p) = (D_{\phi(p)}f)(Y_{\phi(p)}) = (D_{\phi(p)}f)(D_p\phi)(X_p) \tag{1.1}$$
$$= ((D_p\phi)X_p)(f) = L_X(f \circ \phi)(p). \tag{1.2}$$

Hence $X'Y'(f\circ\phi) = X'((Yf)\circ\phi) = (XYf)\circ\phi$; similarly, $Y'X'(f\circ\phi) = (YXf)\circ\phi$. Therefore,

$$[X', Y'](f \circ \phi) = X'Y'(f \circ \phi) - Y'X'(f \circ \phi) \tag{1.3}$$
$$= (XYf) \circ \phi - (YXf) \circ \phi \tag{1.4}$$
$$= ([X, Y]f) \circ \phi. \tag{1.5}$$

Question 1.2.2. *Show that*

- $\mathrm{GL}(n, \mathbb{R})$ *is a non-compact Lie group of n^2 dimension;*
- $\mathrm{GL}(n, \mathbb{C})$ *is a non-compact Lie group of $2n^2$ dimension;*

- *both of them are non-abelian for* $n \geq 2$.

Solution:

Let $M(n,\mathbb{R})$ denote the set of $n \times n$ matrices with real entries. Because it is a real vector space of dimension n^2 under matrix addition and scalar multiplication, $M(n,\mathbb{R})$ is a smooth n^2-dimensional smooth manifold. Now $\mathrm{GL}(n,\mathbb{R})$ is an open subset of $M(n,\mathbb{R})$, namely the set where $\det : M(n,\mathbb{R}) \to \mathbb{R}$ is nonzero, hence $\mathrm{GL}(n,\mathbb{R})$ is a smooth n^2-dimensional submanifold of $M(n,\mathbb{R})$. Moreover, for all $x \in \mathrm{GL}(n,\mathbb{R})$, $\lambda x \in \mathrm{GL}(n,\mathbb{R}) \quad (\lambda \in \mathbb{R} - \{0\})$. Therefore $\mathrm{GL}(n,\mathbb{R})$ is not bounded, hence non-compact.

Next we prove that $\mathrm{GL}(n,\mathbb{R})$ admits a Lie group structure. Multiplication is smooth because the matrix entries of a product matrix AB are polynomials in the entries of A and B. Inversion is smooth by Cramer's Rule. When $n \geq 2$, $AB \neq BA$, hence $\mathrm{GL}(n,\mathbb{R})$ is not Abelian when $n \geq 2$.

The case for $\mathrm{GL}(n,\mathbb{C})$ is similar.

Question 1.2.3. *Find an explicit Lie group homomorphism*

$$\phi : SU(3) \times SU(2) \times U(1) \to SU(5)$$

with discrete kernel.

Solution:

Consider the homomorphism

$$\phi : SU(3) \times SU(2) \times U(1) \to SU(5) \tag{1.6}$$

$$(U_3, U_2, e^{i\theta}) \mapsto \begin{pmatrix} e^{-2i\theta}U_3 & 0 \\ 0 & e^{3i\theta}U_2 \end{pmatrix}. \tag{1.7}$$

- ϕ is indeed a Lie group homomorphism. ϕ is clearly a group homomorphism; to prove ϕ is smooth, note just

  $$\phi(U_3, U_2, e^{i\theta}) = \mathrm{diag}\{e^{-2i\theta}, e^{-2i\theta}, e^{-2i\theta}, e^{3i\theta}, e^{3i\theta}\} \begin{pmatrix} U_3 & 0 \\ 0 & U_2 \end{pmatrix}.$$

  The two terms of the left hand side of the above equation are embeddings of $U(1)$ and $SU(3) \times SU(2)$ into $SU(5)$, respectively. Hence they are smooth; but also product operation in Lie group is smooth, so ϕ is smooth.
- The kernel of ϕ is discrete.

  If $(U_3, U_2, e^{i\theta}) \in \mathrm{Ker}\,\phi$, then $U_3 = \mathrm{diag}\{e^{2i\theta}, e^{2i\theta}, e^{2i\theta}\}$, $U_2 = \mathrm{diag}\{e^{-3i\theta}, e^{-3i\theta}\}$. Moreover, $U_3 \in SU(3)$, $U_2 \in SU(2)$ implies $e^{6i\theta} = 1$, which means that the possible values of θ are taken from $\left\{\frac{n\pi}{3} : n \in \mathbb{Z}\right\}$, which is discrete.

Question 1.2.4. *Let* $\mathbb{K} = \mathbb{R}, \mathbb{C}$ *and identify the Lie algebra of an embedded Lie subgroup of* $\mathrm{GL}(n,\mathbb{K})$ *with a Lie subalgebra of* $\mathrm{Mat}(n \times n, \mathbb{K})$. *Prove that*

- *The Lie bracket on the Lie subalgebra of* $\mathrm{Mat}(n \times n, \mathbb{K})$ *is the standard commutator of matrices.*
- *The Lie algebras of classical groups are the following Lie subalgebras of* $\mathrm{Mat}(n \times n, \mathbb{K})$.

$$\mathfrak{sl}(n,\mathbb{K}) = \{M \in \mathrm{Mat}(n \times n, \mathbb{K}) : \mathrm{tr}(M) = 0\},$$

$$\mathfrak{o}(n) = \{M \in \text{Mat}(n \times n, \mathbb{K}) : M + M^T = 0\},$$

$$\mathfrak{u}(n) = \{M \in \text{Mat}(n \times n, \mathbb{K}) : M + M^\dagger = 0\},$$

$$\mathfrak{so}(n) = \mathfrak{o}(n),$$

$$\mathfrak{su}(n) = \{M \in \text{Mat}(n \times n, \mathbb{K}) : M + M^\dagger = 0, \text{tr}(M) = 0\}.$$

[Hint : Use the fact that 'If H is an embedded Lie subgroup of G, then $\mathfrak{h}$ is an embedded Lie subalgebra of $\mathfrak{g}$' and compute the dimension of each matrix group.]

See [Ha17] Theorem 1.5.22 and 1.5.27.

Question 1.2.5. *Let $X \in \mathfrak{g}$ and $g \in G$ with $(G, \mathfrak{g})$ lying in*

$$\{(SL(n,\mathbb{K}), \mathfrak{sl}(n,\mathbb{K})), (O(n), \mathfrak{o}(n)), (U(n), \mathfrak{u}(n)), (SO(n), \mathfrak{so}(n)), (SU(n), \mathfrak{su}(n))\}.$$

Show that $\mathfrak{g}$ is closed under the map

$$X \mapsto g \cdot X \cdot g^{-1}.$$

Solution: Use Question 1.2.4 and DIRECT calculation.

Question 1.2.6. *Let G be a connected Lie group. Define the centre of G as*

$$Z(G) := \{g \in G : gh = hg, \forall h \in G\}.$$

Prove that

- *$Z(G)$ is the kernel of the adjoint representation Ad_G.*
- *$Z(G)$ is an embedded Lie subgroup in G with Lie algebra given by the centre $\mathfrak{z}(\mathfrak{g})$ of $\mathfrak{g}$.*
- *$\mathfrak{g}$ is abelian if and only if G is abelian.*
- *Ad_G is trivial (i.e. every element of G is mapped to the identity) if and only if G is abelian.*
- *The left-invariant and right-invariant vector fields on a connected Lie group G coincide if and only if G is abelian.*

Solution: Suppose that Ad_g is trivial, and let $h = \exp(v) \in G$. Now $\gamma(t) = g \cdot \exp \cdot g^{-1}$ is a curve in G whose velocity at $t = 0$ is $\text{Ad}_g(v) = v$, since Ad_g is trivial. However, $\rho(t) = \exp(tv)$ is also a curve in G with initial velocity v, and by the uniqueness of integral curve, $g \cdot \exp(tv) \cdot g^{-1} = \exp(tv)$. In particular, when $t = 1$ one has $gh = hg$. For general $h \in G$, since G is connected, by Proposition 7.14 of [Lee13], h can always be written as $h = \exp(v_1) \cdots \exp(v_k)$, and the commutativity comes from iterating what we have done above. Hence $\text{Ker}(\text{Ad}_g) \subset Z(G)$. On the other hand elements of $Z(G)$ clearly fall into $\text{Ker}(\text{Ad}_G)$. Therefore $\text{Ker}(\text{Ad}_G) = Z(G)$.

Now by Proposition 20.8 of [Lee13], we have a commutative diagram

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\text{ad}} & \text{End}(\mathfrak{g}) \\ \Big\downarrow{\scriptstyle \exp} & & \Big\downarrow{\scriptstyle \exp} \\ G & \xrightarrow{\text{Ad}} & \text{GL}(\mathfrak{g}) \end{array}$$

Since $\text{Ker}(\text{Ad}_G) = Z(G)$, the Lie algebra $\mathfrak{z}(\mathfrak{g})$ of $Z(G)$ is given by $\text{Ker}(\text{ad}_{\mathfrak{g}}) = \{X \in \mathfrak{g} : [X, Y] = 0, \forall Y \in \mathfrak{g}\}$. In order this correspondence behaves well, one must demonstrate that $Z(G)$ is a closed subgroup of G. For this, let $C_g = \{g_1 \in G : g_1 g = g g_1\}$. Then C_g is the preimage $f^{-1}(\{e\})$ where f is the smooth map $f(g_1) := g g_1 g^{-1} g_1^{-1}$. Therefore, C_g is closed, and then $Z(G) = \cap_{g \in G} C_g$ is a closed subgroup of G.

Reference

参考文献

[Lee13] John M. Lee. *Introduction to Smooth Manifolds*. Graduate Texts in Mathematics. Springer, 2013.

[Ha17] Hamilton, Mark J.D. *Mathematical Gauge Theory*. Universitext. Springer, 2017.

习题: 规范场/Example Sheet 2

原网页

Question 1.2.1. *Consider a group G and a set M.*

- *Let*

$$G \times M \to M \tag{1.8}$$
$$(g, p) \mapsto g \cdot p \tag{1.9}$$

be a left action. Then

$$M \times G \to M \tag{1.10}$$
$$(p, g) \mapsto p * g = g^{-1} \cdot p \tag{1.11}$$

defines a right action of G on M.

- *Find an example of a left action*

$$G \times M \to M \tag{1.12}$$
$$(p, g) \mapsto p * g = g \cdot p \tag{1.13}$$

does not *define a right action of G on M.*

Question 1.2.2. *Consider the left action of $SU(2)$ on $\mathbb{C}^2$, defined by matrix multiplication,*

$$SU(2) \times \mathbb{C}^2 \to \mathbb{C}^2 \tag{1.14}$$
$$(g, v) \mapsto gv \tag{1.15}$$

and vectors

$$\tau_a = -\frac{\iota \sigma_a}{2} \in \mathfrak{su}(2), \quad a = 1, 2, 3.$$

- *Determine the fundamental vector fields $\tilde{\tau}_a$ on $\mathbb{C}^2$.*

- *Show,* by direct calculation*, that*

$$[\tilde{\tau}_a, \tilde{\tau}_b] = \widetilde{[\tau_a, \tau_b]}, \quad \forall a, b \in \{1, 2, 3\}.$$

- Calculate directly $l_{A*}(\tilde{\tau}_1)$ *with*

$$A = \begin{pmatrix} r & -\bar{r} \\ r & \bar{r} \end{pmatrix} \in SU(2), \quad r = \frac{1}{2} - \frac{1}{2}\iota.$$

- *Compare* $l_{A*}(\tilde{\tau}_1)$ *with* $\widetilde{\mathrm{Ad}_A \tau_1}$.

Question 1.2.3. *Let G, H be Lie groups with a Lie group homomorphism $\phi : G \to H$. Suppose that G acts continuously on the right on smooth manifold M and H acts continuously on the right on smooth manifold N. Consider a ϕ-equivariant map f, i.e.*

$$f(p \cdot g) = f(p) \cdot \phi(g), \quad \forall p \in M, g \in G.$$

Prove

- *If f is continuous, then f induces a continuous map*

$$f_\phi : M/G \to N/H.$$

- *If ϕ is an isomorphism and f is a homeomorphism, then f_ϕ is a homeomorphism.*

Question 1.2.4. *Consider the half-plane*

$$H = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$$

and the special linear group

$$SL(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : ad - bc = 1 \text{ with } a, b, c, d \in \mathbb{R} \right\}$$

- *Show that the map*

$$SL(2, \mathbb{R}) \times H \to H \tag{1.16}$$

$$(A, z) \mapsto \frac{az + b}{cz + d} \tag{1.17}$$

  is a well-defined left-action of $SL(2, \mathbb{R})$ on H.
- *Prove that this action is transitive and that it defines a diffeomorphism between H and $SL(2, \mathbb{R})/SO(2)$.*

Question 1.2.5. *Consider the Möbius strip*

$$M = \left\{ \left(e^{\iota\phi}, re^{\iota\phi/2} \right) : \phi \in [0, 2\pi], r \in [-1, 1] \right\} \subset \mathbb{S}^1 \times \mathbb{C}$$

Define the projection $\pi : M \to \mathbb{S}^1$ by $\pi = \mathrm{pr}_1 |_M : \mathbb{S}^1 \times \mathbb{C} \to \mathbb{S}^1$. Denote the map $f_n : \mathbb{S}^1 \to \mathbb{S}^1$, $f_n(z) = z^n$ for $n \in \mathbb{Z}$.

- *Show that $\pi : M \to \mathbb{S}^1$ is a fibre bundle with general fibre $[-1, 1]$.*

- *Prove that the boundary ∂M is connected and that the bundle $\pi : M \to \mathbb{S}^1$ is not trivial.*
- *Prove that the image of any smooth section $s : \mathbb{S}^1 \to M$ intersects the zero section $z : \mathbb{S}^1 \to M$, $z(\alpha) = (\alpha, 0)$.*
- *Show that the pull-back bundle $f_n^* M$ is isomorphic to the bundle $M_n \to \mathbb{S}^1$ given by*

$$M_n := \left\{ \left(e^{\iota\psi}, re^{\iota n\psi/2} \right) : \psi \in [0, 2\pi], r \in [-1, 1] \right\} \subset \mathbb{S}^1 \times \mathbb{C}.$$

- *Determine those $n \in \mathbb{Z}$ for which $f_n^* M$ is trivial and those for which it is non-trivial.*

Question 1.2.6. *Let $G \to P \xrightarrow{\pi} M$ be a principal bundle and $f : N \to M$ a smooth map between smooth manifolds. Prove that the pullback f^*P has the canonical structure of a principal G-bundle over N.*

Question 1.2.7. *Let $F \to E \xrightarrow{\pi} M$ and $F' \to E' \xrightarrow{\pi} M$ be fibre bundles and $H : E \to E'$ a bundle morphism. Suppose that H maps every fibre of E diffeomorphically onto a fibre of E'. Show that H is a bundle isomorphism.*

Question 1.2.8. *Let $G \to P \xrightarrow{\pi} M$ and $G' \to P' \xrightarrow{\pi} M$ be principle bundles. Suppose $f : G \to G'$ is a Lie group isomorphism. Show that every f-equivariant bundle morphism $H : P \to P'$ is a diffeomorphism.*

Question 1.2.9. *Let $E \to M$ be a $\mathbb{K}$-vector bundle of rank m with a positive definite bundle metric. Suppose $F \subset E$ is a vector subbundle. Prove that the orthogonal complement $F^\perp$ is a vector subbundle of E and that $F \oplus F^\perp$ is isomorphic to E.*

Question 1.2.10. *Let $G \to P \xrightarrow{\pi} M$ be a principle bundle and $\rho : G \to \mathrm{GL}(V)$, $\rho_i : G \to \mathrm{GL}(V_i)$ for $i = 1, 2$ representations on V, V_i. Consider the associated vector bundles*

$$E = P \times_\rho V, \quad E_i \times_{\rho_i} V_i.$$

- *Show that the dual bundle E^*, the direct sum $E_1 \oplus E_2$ and the tensor product $E_1 \otimes E_2$ are isomorphic to vector bundles associated to P.*
- *Determine the corresponding representations of G and the vector bundle isomorphisms.*

Question 1.2.11. *Let $E \to M$ be a complex vector bundle of rank $n \geq 2$. Show that E is associated to a principal $SU(n)$-bundle over M if and only if $\Lambda^n E$ is a trivial complex line bundle.*

Question 1.2.12. *Let $E = P \times_\rho V$ be an associated vector bundle and α a section of the adjoint bundle $\mathrm{Ad}(P)$. Prove that α defines a canonical endomorphism of the vector bundle E.*

习题: 规范场/Example Sheet 3

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Question 1.2.1. *Let G be a Lie group and $x : [0, 1] \to \mathfrak{g}$ a smooth map. Denote by $X(t)$ the associated left-invariant or right-invariant time-dependent vector field on G. Prove that there exists a unique smooth integral curve $g : [0, 1] \to G$ such that*

$$\dot{g}(t) = X(t) \qquad \forall t \in [0, 1]$$
$$g(0) = e.$$

Question 1.2.2. *Let $(E, \pi, M; F)$ be a fibre bundle. Prove that the bundle is isomorphic to a trivial bundle if and only if there exists a smooth map $\tau : E \to F$ such that the restrictions*

$$\tau|_{E_x} : E_x \longrightarrow F$$

are diffeomorphisms for all $x \in M$.

Question 1.2.3. *Prove that the vertical tangent space and vertical tangent bundle satisfy*

- *$V_p = \operatorname{Ker} D_p\pi$;*
- *The map*

$$\begin{aligned} \phi_* : \mathfrak{g} &\longrightarrow V_p \\ X &\longmapsto \tilde{X}_p \end{aligned}$$

  is a vector space isomorphism, where $\tilde{X}$ is the fundamental vector field given by the G-action on P;
- *V has a rank equal to $\dim G$ and a global trivialization*

$$\begin{aligned} P \times \mathfrak{g} &\longrightarrow V \\ (p, X) &\longmapsto \tilde{X}_p; \end{aligned}$$

- *V is right-invariant, i.e.*

$$r_{g_*}(V_p) = V_{p\cdot g}, \quad \forall g \in G.$$

Question 1.2.4. *Let $P \to M$ be a principal bundle and $A \in \Omega^1(P, \mathfrak{g})$ a connection 1-form on P, $F \in \Omega^2(P, \mathfrak{g})$ its curvature 2-form, $f \in \mathscr{G}(P)$ a global bundle automorphism, X and Y horizontal vector fields on P with respect to the Ehresmann connection H determined by A.*

- *Prove that f^*A is a connection 1-form on P and*

$$f^*A = \operatorname{Ad}_{\sigma_f^{-1}} \circ A + \sigma_f^* \mu_G.$$

- *Prove that*

$$F(X, Y) = -A([X, Y]).$$

Question 1.2.5. *Let $P \to M$ be a principal G-bundle with a matrix Lie group G, E an associated vector bundle, A a connection 1-form on P. Choose an open set $U \subset M$ such that P_U is trivial and set a local gauge $s : U \to P$ and a loop $\gamma \subset U$. Prove that*

$$\operatorname{Hol}^E_{\gamma,x}(A) = \rho\left(\mathscr{P}\exp\left(-\oint_\gamma A_s\right)\right) = \mathscr{P}\exp\left(-\oint_\gamma \rho_* A_s\right),$$

with respect to the isomorphism

$$\begin{aligned} V &\longrightarrow E_x \\ v &\longmapsto [s(x), v]. \end{aligned}$$

Question 1.2.6. *Let $P \to M$ be a principal G-bundle, A a connection 1-form, F the curvature 2-form of A, H the horizontal subbundle determined by A.*

- *Prove that dF vanishes on $H \times H \times H$.*
- *Prove the identity*

$$dF + [A, F] = 0.$$

Question 1.2.7. *Let (M, g) be an oriented pseudo-Riemannian manifold of signature (s, t) and $*$ the Hodge star operator. Prove that*

$$** = (-1)^{t+k(n-k)}.$$

Question 1.2.8. *Let (M, g) be a pseudo-Riemannian 4-manifold and $P \to M$ a principal bundle. Prove that the Yang-Mills action is invariant under a conformal change of the metric g, that is*

$$g' = e^{2\lambda} g$$

for any $\lambda \in C^\infty(M)$.

Question 1.2.9. *The Higgs Lagrangian of scalar fields is gauge invariant. That is,*

$$\mathscr{L}_{\mathrm{H}}\left[f^{-1}\Phi, f^*A\right] = \mathscr{L}_{\mathrm{H}}[\Phi, A], \quad \forall f \in \mathscr{G}(P).$$

Question 1.2.10. *Fix the following geometry.*

- *(M, g) : the Minkowski spacetime with signature $(+, -, -, -)$,*
- *G : a compact Lie group with a Lie algebra $\mathfrak{g}$,*
- *$P \to M$: a principal G-bundle,*
- *$\mathrm{Ad}(P)$: the adjoint bundle with bundle metric $\langle \cdot, \cdot \rangle_{\mathrm{Ad}(P)}$,*
- *$E = P \times_\rho \mathbb{C}^n \to M$: an associated vector bundle with representation ρ and G-invariant Hermitian bundle metric $\langle \cdot, \cdot \rangle_E$,*
- *A : connection 1-forms on P,*
- *F_A : curvature 2-forms on P with respect to A,*
- *Φ : smooth sections of E,*
- *d_A : the exterior covariant derivative on the adjoint bundle $\mathrm{Ad}(P)$,*
- *$d_{A,\rho}$: the exterior covariant derivative on the associated vector bundle E with representation ρ,*

$$d_{A,\rho} : \Omega^k(M, E) \longrightarrow \Omega^{k+1}(M, E).$$

The Yang-Mills-Klein-Gordon Lagrangian is defined to be

$$\mathscr{L}_{\mathrm{KG}}[\Phi, A] + \mathscr{L}_{\mathrm{YM}}[A] = \langle d_{A,\rho}\Phi, d_{A,\rho}\Phi \rangle_E - m^2 \langle \Phi, \Phi \rangle_E - \frac{1}{2} \left\langle F_M^A, F_M^A \right\rangle_{\mathrm{Ad}(P)}.$$

Derive the coupled system of Yang-Mills-Klein-Gordon equations.

试卷: 规范场

原网页

Example Sheet

Question 1. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and identify the Lie algebra of an embedded Lie subgroup of $\mathrm{GL}(n, \mathbb{K})$ with a Lie subalgebra of $\mathrm{Mat}(n \times n, \mathbb{K})$. Prove that

- The Lie bracket on the Lie subalgebra of $\mathrm{Mat}(n \times n, \mathbb{K})$ is the standard commutator of matrices.
- The Lie algebras of classical groups are the following Lie subalgebras of $\mathrm{Mat}(n \times n, \mathbb{K})$.
  - The Lie algebras of the special linear groups $\mathrm{SL}(n, \mathbb{K})$ read

$$\mathfrak{sl}(n, \mathbb{K}) = \{M \in \mathrm{Mat}(n \times n, \mathbb{K}) : \mathrm{tr}(M) = 0\}.$$

  - The Lie algebras of the orthogonal groups $\mathrm{O}(n)$ read

$$\mathfrak{o}(n) = \left\{M \in \mathrm{Mat}(n \times n, \mathbb{R}) : M + M^T = 0\right\}.$$

  - The Lie algebras of the unitary groups $\mathrm{U}(n)$ read

$$\mathfrak{u}(n) = \left\{M \in \mathrm{Mat}(n \times n, \mathbb{C}) : M + M^\dagger = 0\right\}.$$

  - The Lie algebras of the special orthogonal groups $\mathrm{SO}(n)$ read

$$\mathfrak{so}(n) = \mathfrak{o}(n).$$

  - The Lie algebras of the special unitary groups $\mathrm{SU}(n)$ read

$$\mathfrak{su}(n) = \left\{M \in \mathrm{Mat}(n \times n, \mathbb{C}) : M + M^\dagger = 0, \mathrm{tr}(M) = 0\right\}.$$

[Hint: Use the fact that 'If H is an embedded Lie subgroup of G, then $\mathfrak{h}$ is an embedded Lie subalgebra of $\mathfrak{g}$' and compute the dimension of each matrix group.]

Question 2. Let $(E, \pi, M; F)$ be a fibre bundle. Prove that the bundle is isomorphic to a trivial bundle if and only if there exists a smooth map $\tau : E \to F$ such that the restrictions

$$\tau|_{E_x} : E_x \longrightarrow F$$

are diffeomorphisms for all $x \in M$.

Question 3. Let $s_i : U_i \to P$ and $s_j : U_j \to P$ be local gauges with $U_i \cap U_j \neq \emptyset$ and $g_{ij} : U_i \cap U_j \to G$ transition functions. Prove that the local curvature 2-forms F_{s_i}, F_{s_j} transform as

$$F_{s_i} = \mathrm{Ad}_{g_{ji}^{-1}} \circ F_{s_j} \quad \text{ on } U_i \cap U_j.$$

In particular, when G is a matrix Lie group, prove that

$$F_{s_i} = g_{ji}^{-1} \cdot F_{s_j} \cdot g_{ji}.$$

[Hint: Adapt the proof of the analogous theorem for local connection 1-forms.]

Question 4. Show that the exterior covariant derivative d_A satisfies

$$\begin{aligned} d_A(\omega + \omega') &= d_A\omega + d_A\omega' && \forall \omega, \omega' \in \Omega^k(M, E) \\ d_A(\sigma \otimes e) &= d\sigma \otimes e + (-1)^k \sigma \wedge \nabla^A e && \forall \sigma \in \Omega^k(M), e \in \Gamma(E) \\ d_A(\sigma \wedge \omega) &= d\sigma \wedge \omega + (-1)^k \sigma \wedge d_A\omega && \forall \sigma \in \Omega^k(M), \omega \in \Omega^l(M, E). \end{aligned}$$

[Hint: Invoke the local expression of the exterior covariant derivative.]

Question 5. We live in the following geometry.

- $(\mathbb{R}^4, g)$: the Minkowski spacetime with signature $(+,-,-,-)$,
- G: a compact Lie group (say $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$) with a Lie algebra $\mathfrak{g}$,
- $P \to M$: a principal G-bundle,
- $\mathrm{Ad}(P)$: the adjoint bundle with bundle metric $\langle \cdot, \cdot \rangle_{\mathrm{Ad}(P)}$,
- $E \to M$: an associated vector bundle with representation ρ and Hermitian bundle metric $\langle \cdot, \cdot \rangle_E$.

Denote

- A: connection 1-forms on P,
- F_A: curvature 2-forms on P with respect to A,
- Φ: smooth sections of E,
- d_A: the exterior covariant derivative on the adjoint bundle $\mathrm{Ad}(P)$,
- $d_{A,\rho}$: the exterior covariant derivative on the associated vector bundle E with representation ρ,

$$d_{A,\rho} : \Omega^k(M, E) \longrightarrow \Omega^{k+1}(M, E).$$

Consider the Yang–Mills–Higgs Lagrangian

$$\mathscr{L}_{\mathrm{YMH}}[\Phi, A] = \langle d_{A,\rho}\Phi, d_{A,\rho}\Phi \rangle_E - V(\langle \Phi, \Phi \rangle_E) - \frac{1}{2}\langle F_A, F_A \rangle_{\mathrm{Ad}(P)}$$

with a function $V(x) = x^2/2 - 1$ on $\mathbb{R}$.

- Prove that variation of Φ leads to the field equation

$$d^*_{A,\rho} d_{A,\rho} \Phi = V'(\langle \Phi, \Phi \rangle_E). \tag{1}$$

- Prove that variation of A leads to the field equation

$$d^*_A F_A \Phi = J_\rho(A, \Phi),$$

where $J_\rho(A, \Phi) \in \Omega^1(\mathbb{R}^4, \mathrm{Ad}(P))$ is a twisted 1-form defined by

$$\langle \alpha, J_\rho(A, \Phi) \rangle_{\mathrm{Ad}(P)} = 2\Re\left(\langle d_{A,\rho}\Phi, \alpha \cdot \Phi \rangle_E\right), \quad \forall \alpha \in \Omega^1(\mathbb{R}^4, \mathrm{Ad}(P)). \tag{2}$$

[Background in physics: The system (1)-(2) is called the **Yang–Mills–Higgs equation**. It is used in Higgs mechanism to explain how gauge bosons A gain mass through interactions with Higgs bosons Φ. The Higgs bosons were proposed in 1960s and finally detected at CERN in 2012.]